DGPF-RENet: A Low Data Dependence Network With Low Training Iterations for Hyperspectral Image Classification

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***Abstract*— The classification of ground objects from hyperspectral images (HSIs) is of great importance for human perception of information about the terrain and landscape. HSIs have numerous dimensions, and obtaining the data is difficult. The issue of slow convergence of neural network training is brought on by high-dimensional data, and the neural network's performance is impacted by the challenging data acquisition process. In order to achieve the effects of low data dependence and rapid convergence, we propose a redundancy elimination network architecture with a decoupled-gaze attention mechanism and phantom fractal modules (DGPF-RENet) for HSI classification. First, we propose the decoupled-gaze attention (DGA) mechanism to make full use of the correlation between adjacent bands and the continuity of neighboring pixels in HSIs. Then, a redundancy elimination module (REM) is proposed to reduce the number of feature points and eliminate redundant information while preserving the contextual information and relationships between pixels. Finally, the phantom fractal module (PFM) is proposed, which improves the scale of feature learning by fractalizing convolutions at multiple scales. Four publicly available HSIs datasets, including Indian Pines, Salinas, DFC2018, and WHUHi-HongHu, were used in our experiments. According to experimental findings, compared to other state-of-the-art methods, our method performs best with a small number of training samples and few iterations. We have released our code and models at https://github.com/yuhua666/DGPF-RENet.**

***Index Terms*— Attention, eliminate redundancy, feature learning, hyperspectral images (HSIs), image classification.**

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I. Introduction

REMOTE sensing images have a wide range of applications in various fields due to their high information content [1]. The rapid development of spectral imaging technology [2] has already brought a promising future for the development of remote sensing technology [3]. Hyperspectral remote sensing, as a remote sensing technique, can acquire rich spatial and spectral information in the target area at the same time and has become one of the important means of ground observation [4]. Different from multispectral remote sensing, hyperspectral remote sensing can acquire information in hundreds of continuous spectral wave bands of ground objects, providing rich spectral information to enhance the ability to distinguish between ground objects [5]. The classification of ground objects from hyperspectral images (HSIs) is an important tool for cognitive topography and other information, and it is important for a variety of fields, such as precision agriculture [6], environmental monitoring, and geological surveys. In recent years, the study of ground object classification by hyperspectral remote sensing has become popular among researchers and government agencies, and a variety of aerial hyperspectral sensors and satellite-based hyperspectral sensors are now being used for ground observation applications [7].

Most early approaches to HSI classification studies concentrate on the role of spectral characteristics by employing conventional pixel classification techniques. For the categorization of HSI, for instance, multilayer perceptrons [8], support vector machines (SVMs) [9], polynomial logistic regression [10], and dynamic or random subspace [11] have been suggested. However, the classification maps produced by the aforementioned method do not take into account the spatial context, which makes it difficult to recover the actual location data for ground items. A spectral–spatial feature-based classification (SSFC) framework, which incorporates spatial context information into the pixel classification method, is gradually investigated as a result of the realization of the significance of spatial dimension features in restoring the true relative positions of objects [12]. Extended morphological contour (EMP) [13] is used to utilize spatial information through multiple morphological operations and multicore learning to explore the spectral–spatial information of HSIs because hyperspectral pixels can typically be represented by a linear combination of

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several common pixels from the same class. A postprocessing method called edge retention filtering (EPF) [14] is used to enhance the probabilistic outcomes of SVM. Furthermore, some techniques study spatial consistency in HSIs by subdividing them into several superpixels based on texture or intensity similarity [15], [16], [17], and techniques such as band selection and low-rank decomposition enable the straightforward and efficient application of traditional handcrafted descriptors in specific scenarios [18]. However, the limitations of handcrafted features arise from their task-specific nature and reliance on expert knowledge for parameter setting. These features struggle to effectively capture and represent the complexity present in large datasets, thereby limiting their applicability in challenging real-world scenarios. In the context of HSI categorization, the extraction of highly discriminative features is of utmost importance. Manual characteristics may not possess the necessary representational capacity to identify subtle differences between categories or significant variations within the same category.

Deep learning architecture had considerable success with natural picture identification problems over the past decade [19]. It has a significant capacity for learning data features and the ability to distinguish between various items. Many studies use it to classify HSIs and use deep learning techniques to extract richer HSIs properties. When learning tens or hundreds of layers of neural networks, deep learning technology can extract a significant number of information features from the original data compared to typical human approaches. The deep belief network (DBN), the recurrent neural network (RNN), and the convolutional neural network (CNN) are the three primary types of common deep learning models for HSI categorization. The RNN-based HSI classification approach suffers from substantial gradient disappearance issues due to strong order reliance on spectral bands. This makes it challenging to learn long-term correlation and causes performance bottlenecks in practical HSI classification. For DBN, the deep learning algorithm cannot estimate the law of data without bias in the application scenarios that can only supply a limited amount of data due to the random initialization of weight parameters. As a result, it is challenging to complete the hyperspectral picture training assignment with few samples, and overfitting is simple to produce. In comparison to DBN and RNN, CNN is effective in extracting information about the spatial structure of images. It is the most extensively used deep learning network model in the field of HSI classification and is applied to a variety of tasks in natural situations, including image classification, target recognition, scene segmentation, and others.

Compared with RGB images, HSIs usually have higher dimensionality. In the practical process of HSI classification tasks, there are three main problems that need to be solved.

1) HSIs have high-dimensional information, and there is a strong correlation between adjacent spectral bands; some image fusion methods [20], [21] have demonstrated the importance of considering the joint consideration of spectral information and spatial information. However, most previous methods [22], [23] for HSI classification take labeled high-dimensional spectral pixels as input to the network and generate pixel blocks by extracting a certain size of the neighborhood around each pixel. This approach often overlooks the global context information and interband correlation, resulting in many isolated salt-and-pepper noise points in the classification results.

2) Compared to the nanometer-level resolution of infrared spectra, the spatial resolution of high-spectral images is relatively low. These images often contain mixed-color pixels, which include a significant amount of redundant information that can adversely affect the discriminative ability of the models.

3) HSIs contain richer local information, such as scene structures, object shapes, and spatial distributions, compared to natural optical images. Therefore, the design of scale information in the feature extraction process needs to be fully considered to enhance the extraction of local information.

To explore the combination of spectral and spatial features of HSIs, Al-Khafaji et al. [24] proposed a method to explore both spectral and spatial dimensions to extract spectral features and geometrically transform invariant features, detected from extrema after applying a 3-D difference of Gaussian to data cube. Ahmad et al. [25] proposed a gray-scale co-occurrence matrix computational morphological descriptor to determine unique spatial features, aiming to combine spectral and spatial information. However, these methods only combine spectral and spatial information without proper handling of the correlation between adjacent bands in the spectral dimension and the continuity of neighboring pixels in the spatial dimension. These important pieces of information are easily overlooked, leading to the occurrence of salt-and-pepper noise.

To eliminate redundant information in HSIs, Vasin et al. [26] proposed a local, homogeneous "well-adapted" basis function, reducing the interference of redundant information with the HSI feature learning process. Liu et al. [27] proposed a new hybrid dilation convolution-guided feature filtering and enhancement strategy, introducing atrous convolution to reduce spatial feature loss without reducing the size of the receptive field. While these methods eliminate redundant information on HSIs, they also cause a loss of interpixel contextual information, which can easily lead to discontinuous classification results [28].

To enhance the scale information of the model, Zhu et al. [29] proposed a multiscale short- and long-range graph convolutional network (MSLGCN) for HSI classification. The combination of long- and short-range graph convolutions utilizes rich and complementary multiscale information to extract spatial information of different scales, enabling the model to learn different-sized texture structures and obtain fine boundaries. Ding et al. [30] introduced a novel multiscale receptive field graph attention neural network that abstracts the local spatial features of HSIs using a superpixel segmentation method. They employed a 1-D neural network spectral transformation mechanism to extract spectral features of superpixels and then designed a novel multiscale receptive field graph attention network to capture local–global neighboring node features and edge features. Despite the enhancement of scale information in the abovementioned methods, they rely on high computational complexity and training data fitting of

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numerous multiscale graph neural networks, resulting in lower model efficiency and data efficiency.

In summary, this article proposes a redundancy elimination network architecture with a decoupled-gaze attention mechanism and phantom fractal modules (DGPF-RENet). By leveraging spatial–spectral features and enhancing scale information, the architecture reduces the model's data dependence. It achieves accurate classification results under low data dependence and low training iterations by removing redundant information pixel by pixel, accelerating training convergence, and focusing on important information.

The main contributions of this article are given as follows.

1) We propose a decoupled-gaze attention (DGA) mechanism module that integrates spatial and spectral information and utilizes a visual gaze mechanism to focus on the correlation between adjacent bands and the continuity of neighboring pixels, aiming to reduce the model's reliance on data.

2) We propose a redundancy elimination module (REM) that removes redundant information pixel by pixel while preserving contextual information and the continuity of feature relationships.

3) We propose a phantom fractal module (PFM) that integrates convolutions at different scales to obtain a collection of multiscale feature maps, enhancing the model's ability to extract local details in HSIs.

4) We evaluate the performance of the proposed methods on four publicly available HSI classification datasets and comprehensively compare them with state-of-the-art methods, demonstrating the superiority and generalization ability of the proposed methods.

The remainder of this article is organized as follows. Section II presents an overview of the related work on HSI classification. Section III introduces the proposed DGPF-RENet. Section IV provides experimental results on four datasets to demonstrate the effectiveness and generalization capability of DGPF-RENet. Section V discusses the reasons for the success of DGPF-RENet and the limitations of the proposed method. Finally, Section VI summarizes the design principles of this method and concludes this article with some remarks and plans for potential future research directions.

II. Related Work

Early HSI classification primarily relied on traditional feature descriptors, which heavily depended on expert knowledge and manual design. With the joint driving force of computational power and data availability, deep neural networks exhibit end-to-end capabilities and powerful feature learning abilities, breaking away from the reliance on traditional feature engineering and achieving high accuracy in HSI classification, even in complex backgrounds. HSIs possess numerous spectral bands, and the abundance of information has prompted researchers to consider the allocation of neural network attention to features. Recent studies have explored the role of attention and transformers in HSI classification. In the following sections, we will introduce the relevant existing works.

A. CNN Methods for HSI Classifications

The application of deep learning in digital image processing and pattern recognition has brought new pixel-based methods, which can extract informative and discriminative features by constructing a hierarchical learning framework [31]. Compared to traditional handcrafted descriptors, deep learning methods have significant advantages and provide new perspectives for HSI classification [32]. In deep learning methods, researchers have applied DBNs and recursive neural networks to HSI classification and demonstrated their superiority over traditional methods. However, these methods require preprocessing steps to transform high-dimensional image data into 1-D vectors, which are time-consuming and labor-intensive, and can lead to the loss of spatial and spectral information in HSIs. Benefiting from the success of CNNs in natural optical images, researchers have developed CNN-based HSI classification networks. Consequently, a series of CNN-based HSI classification methods have been proposed. Liu et al. [33] constructed a supervised deep feature extraction model based on 2-D convolutions as basic units using coordinated extraction of zero-spectral features and metric learning methods to extract highly discriminative coordinated spatial–spectral features from images. However, due to the scarcity of hyperspectral training samples and the prevalent issue of data imbalance, the learning process tends to favor more common categories, which affects the classification results of categories with fewer samples. Xu et al. [34] addressed this issue by increasing the number of channels in this novel deep classification network and designing a combination of 3-D octave convolutions and attention mechanisms to fully utilize the coordinated spatial–spectral information in the images. This approach fully exploits the coordinated spatial–spectral information, but the presence of a large number of 3-D octave convolutions significantly increases the computational cost. Zhong et al. [35] proposed an end-to-end spectral residual network with HSIs having a 3-D cubic structure as input data. Its use of the residual structure effectively saves computational costs, but its training time is too long and too data-dependent to ensure classification stability. To further address the problem of long training times, Wang et al. [36] proposed an end-to-end fast dense spectral–spatial convolutional network, which uses convolutional kernels of different sizes to extract spectral and spatial features, but the dimensionality reduction process tends to lose useful feature information and the classification stability is deficient. To make efficient use of spectrum and spatial feature information, prevent information loss during dimensionality reduction, and lessen the model's reliance on hyperspectral data, Yu et al. [37] proposed an architecture for unsupervised hyperspectral feature extraction based on spatially modified variational autoencoder. First, spectral–spatial information is represented by sequence data; second, spatial features are represented from two separate perspectives by stacked LSTM extraction and stacked CNN (local sensing); and third, KL divergence is inserted into the loss function of the DNN architecture. In digital image processing tasks, the main means of improving feature extraction and classification stability is to deepen the level of the network structure, but CNNs with deep layers can suffer from gradient disappearance and overfitting. In addition, as RGB images and HSV images do not have a spectral dimension, CNN methods applied to digital image processing do not incorporate the difference between the spatial and spectral characteristics of HSIs,

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resulting in inaccurate information extraction and processing. In conclusion, although CNN-based HSI processing methods still have certain shortcomings, convolutional operations can effectively use the spatial–spectral coordinated information in the image and facilitate the effective processing of complex background information, which is a favorable means of processing HSIs.

B. Attention Mechanism for HSI Classifications

The mechanism of attention is based on research into human vision. Due to the processing bottleneck in cognitive science, humans preferentially focus on a subset of all information while disregarding other observable information. During the processing of HSIs data using a neural network model, the attention mechanism can be utilized to focus on the most essential information set among a vast number of input data, reduce the attention to other information, and enhance the accuracy and efficiency of processing jobs. Guo et al. [38] proposed a multibranch network based on weight sharing and an attention mechanism that may assign various weights to different bands in hyperspectral pictures, limit the transmission of noise bands in the network, and boost the transmission of classification-relevant bands. To increase classification performance, Chu et al..).)) [39] proposed a novel approach to enhance the classification performance in computer vision by integrating convolutional block attention module (CBAM) and graph regularization into HSI classification models based on three-dimensional CNN (3-D-CNN) and two-dimensional CNN (2-D-CNN). However, the aforementioned strategy does not eliminate the dependence of the deep learning method on the quantity of data. Compared to RGB images, HSIs are more difficult to produce, and HSI classification has always been hampered by inadequate training samples. To eliminate sample dependence, Pu et al. [40] suggested an aggregated attention switching network (AATN) for HSI classification that may be applied to HSI classification tasks without incurring excessive computational expense. Due to the efficacy and practicability of attention-driven features, the proposed system can acquire representation knowledge with minimum training samples and, thus, improve performance results. End-to-end semisupervised spatial–spectral dual-path network (S3DPN) architecture for learning discriminative spatial–spectral features from a small quantity of labeled data, and a large amount of unlabeled data was proposed by Pu et al. [41]. The unsupervised branch of S3DPN can directly extract the depth representation from the unlabeled sample and simultaneously mine the local detail features and global context information of the unlabeled sample using Oct-Conv, unlike many semisupervised deep learning methods that require the generation of pseudotags (cluster tags). Unlike semisupervised and unsupervised design, the method suggested in this study attempts to reduce the sample reliance of the model through the use of efficient attentional design and in-depth investigation of the link between the characteristics of restricted samples.

C. Transformer for HSI Classifications

Unlike CNN, the vision transformer (ViT) is considered to be a specific tokenization of images into nonoverlapping blocks. By learning the correlations between these blocks, ViT is able to perceive long-range feature relationships more accurately than CNNs and has achieved competitive performance in natural image classification tasks. The success of ViT has been extended to HSI classification tasks, where it is used to model long-range dependencies among a large number of spectral bands in HSIs. The application of ViT in natural image classification tasks can be traced back to the pioneering work of Dosovitskiy et al. [42], who introduced ViT. ViT projects each patch as a fixed-length vector into the transformer and uses special tokens for category prediction. However, its significant defect is the lack of induction bias. Some researchers try to improve the performance of the transformer by introducing a convolution structure to obtain inductive bias. For example, Graham et al. [43] proposed LeViT, whose design idea is combined with ConvNet [44], by using convolutional projection instead of fragment projection and adding an additional layer of nonlinearity to the attention module. Heo et al. [45] reexamined the robustness of ViT and proposed RvT, which introduced some learning components, such as the convolutional layer, pooling layer, and more attention heads.

The effectiveness of ViT has prompted researchers to customize transformer classifiers for HSIs. Hong et al. [46] proposed a backbone network based on the transformer, in which the attention mechanism is used to enhance the ability to capture subtle spectral differences in detail. In order to further unleash the potential of ViTs in HSIs, Chen et al. [47] aim at the multiband characteristics and local spectral differences of HSIs. HiT [48] is proposed to solve the HSI classification problem based on a sequential perspective, which consists of an SACP module for capturing local space-spectral fusion information and long-distance spectral information, and a Conv-Permutator with depth direction convolution and point convolution. In addition, Liu et al. [49] proposed HSI-TNT to address the issues of similar spectra, severe fragmentation, and spatial heterogeneity of different land cover in coastal wetlands. They introduced transformers to fuse local and global features, demonstrating the effectiveness of ViT in real-world scenarios for HSI classification and indicating the significant potential of transformers in the context of HSIs. However, there is limited research on transformers specifically for HSI classification, and most studies have focused on the design of transformer architectures for HSIs or enhancing feature extraction capabilities using CNN. The existing work has inspired this article to explore the feasibility of leveraging transformer characteristics and attention mechanisms to enhance CNN, aiming to improve the feature learning capability for HSI classification.

III. DGPF-RENet

A. Overview of DGPF-RENet

The combination of spectral and spatial features is crucial for improving the performance of deep learning-based models in HSI classification [50]. In order to explore the feature differences between different objects in HSIs, our work aims to integrate spectral and spatial dimensions of

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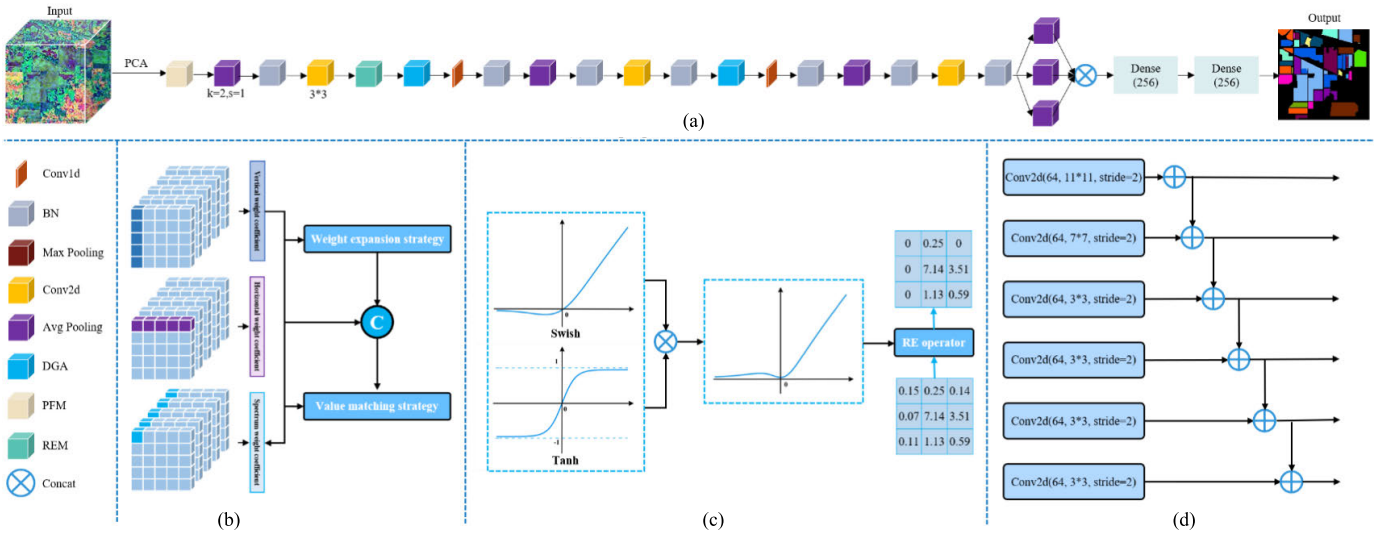


Fig. 1. Structure of DGRE-PFNet. (a) Proposed overall framework. (b) DGA. (c) REM. (d) PFM.

features in HSIs and effectively enhance the model's feature extraction capability to differentiate various land cover classes and prevent category confusion. In addition, we also investigate effective methods to remove redundant and irrelevant information from HSIs, reducing foreground and background classification errors. Based on these design schemes, this article proposes a novel network architecture called DGPF-RENet. The primary goal of DGPF-RENet is to improve the accuracy and generalization ability of HSI recognition while aiming for low data dependence and low training iterations. The DGPF-RENet aims to enhance the data efficiency and model efficiency for the task of HSI classification. Fig. 1 depicts the total DGPF-RENet network design.

DGPF-RENet consists of three modules: DGA, REM, and PFM. The DGA combines spectral features with spatial information and uses visual glance features to mine the relationship between features and improve salt-and-pepper noise in HSI. Given the redundant information present in HSIs, the REM uses a residual structure and a new type of deredundant convolution to reduce redundant information and retain effective features. The REM achieves this by making full use of the contextual relationship of modeled features in HSIs, while the PFM avoids a dramatic increase in the number of parameters by upgrading the feature scale and constructing a multiscale convolution to enhance the feature differentiation capability of the network.

B. Decoupled-Gaze Attention Mechanism

In our previous study, we proposed a module for joint assignment of attention mechanism based on spatial dimensional division, which was obtained with good results in RGB images [51], [52]. However, in HSI recognition tasks, it assigns weights only to spatial dimension and does not pay sufficient attention to important information to achieve good results. The high spatial dimensionality of features and the strong feature correlation of HSIs often lead to an increase in computational cost. Conventional hyperspectral pixel-by-pixel classification methods use only spectral features without considering spatial information, and the classification results often have many discrete isolated points, lowering the classification accuracy and making these methods incompatible with the continuum distribution of the actual ground cover.

Because HSIs' pixels are much smaller and contain finer details than RGB pictures, the previous model is heavily dependent on the amount of data. The human eye is able to distinguish minute details between actual items, including their textures and forms. Human twins, for instance, can be easily distinguished from one another by looking at images of them and observing the small distinctions between them. Inspired by the observation made by the GG-transformer [53], which highlights the ability of the self-attention mechanism and convolution to mimic human glance and gaze mechanisms for holistic environmental perception, we propose the DGA. The DGA leverages the concept of glancing to simulate the focal point selection of human vision. It assigns appropriate weights to spectral and spatial dimensions, enabling the extraction of horizontal and longitudinal spatial continuity, feature similarity between adjacent bands, as well as distinctions between different categories at continuous positions. Feature extraction that focuses on the horizontal and vertical spatial continuity and the similarity of features between adjacent bands can effectively utilize the feature information from limited samples and reduce the model's dependence on data. The strategy is divided into three main parts. The first part is an attention mechanism that assigns decoupled weights in the horizontal, vertical, and spectral dimensions. The second component adds the three values acquired by multiplying two of the three weights in order to loosen the coupling between the three dimensions and improve the gaze effect by increasing the weight coefficient. The third part matches the weighting features and assigns weights to the three types of weights considering the three types of weighting features, which are used to complement the weighting coefficient results obtained in the second part.

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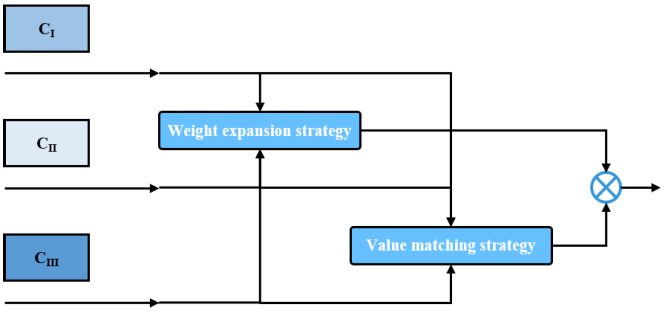


Fig. 2. DGA's structure with its coordinated weight assignment operation. "A" represents the weight expansion strategy. "B" represents the weight-matching strategy. ⊗ indicates the stacking of horizontal, vertical, and spectral dimensional directional weights, expanded weight modulus, and weight-matching strategy.

The DGA aims to take advantage of differences in various dimensions to distinguish differences in detail between objects. The content of the DGA is shown in Fig. 2.

There are three steps in DGA to enlarge the difference in weight coefficient for the features, which are given as follows.

1) Weight Distribution Strategy: We use the attention mechanism to assign gazed weight coefficient dividing into horizontal, vertical, and spectral dimensional orientations

$$c_i = \sum_{j=1}^{n} \frac{\exp(e_{i,j})}{\sum_{k=1}^{n} \exp(e_{i,k})} h_j \tag{1}$$

where $e_{i,j}$ represents the set of pixels in the longitudinal dimension of the space, while $e_{i,k}$ represents the attention weight that captures the interaction between the ith element and the jth element. Similarly, $e_{i,k}$ represents the attention weight for other elements. These attention weights are obtained by applying a softmax function to a set of learnable weights. The softmax function normalizes the exponential values of the weights, resulting in a probability distribution. The numerator $\exp(e_{i,j})$ represents the exponential value of the attention weight $e_{i,j}$, and the denominator is the sum of the exponential weights of all elements, ensuring the normalization of attention scores. Subsequently, these attention weights are used to perform a weighted sum of the corresponding elements h_j, resulting in the final output c_i.

2) Weight Expansion Strategy: We add the three values obtained by multiplying two of the three weights to further expand the weighting factors

$$\text{expand} = (c_I^* c_{II} + c_I^* c_{III} + c_{II}^* c_{III}) \tag{2}$$

where $(c_I = \{c_1, c_2, \ldots, c_{i-1}, c_i\})$ represents the vertical attention mechanism feature sequence, $(c_{II} = \{c_1, c_2, \ldots, c_{i-1}, c_i\})$ represents the horizontal attention mechanism feature sequence, $(c_{III} = \{c_1, c_2, \ldots, c_{i-1}, c_i\})$ represents the attention mechanism feature sequence in the spectral dimension, and expand represents the result of the weight expansion strategy.

3) Value Matching Strategy: To select the maximum value of the 3-D feature information as the main component and to consider the features of the other two dimensions, this strategy matches three types of weight features and is used to replenishment the results of the weight expansion strategy in the previous step

$$\begin{aligned}\text{assignment} = {} & \alpha^* \max(c_I, c_{II}, c_{III}) \\ & + \beta^* \text{median}(c_I, c_{II}, c_{III}) \\ & + \gamma^* \min(c_I, c_{II}, c_{III})\end{aligned} \tag{3}$$

where max represents the maximum value operation, min represents the minimum value operation, median represents the median operation, and assignment represents the value assign strategy. We confirm that the optimal value of the weight assignment parameter is $\alpha = 0.7, \beta = 0.2$, and $\gamma = 0.1$ in Section IV. Notably, we only attempt to alter these three hyperparameters during the experiment's training phase. Once the hyperparameters have been approved, they are fixed and will not be manually altered during the testing phase; hence, the model will not have manual design issues.

The three strategies of the method are combined by the following formula:

$$\text{DGA} = \text{concatenate}([c_I, c_{II}, c_{III}, \text{expand}, \text{assignment}]). \tag{4}$$

C. Redundancy Elimination Module

HSIs usually contain numerous channels, each with feature points that include both meaningful semantic information and redundant data. Excessive redundancy can impact other features. Conventional approaches often overlook the contextual relationships among pixels in HSI processing, leading to noisy and ambiguous classifications. This problem can be mitigated by reducing redundant and irrelevant information in HSIs. The crucial aspect is to extract essential features, such as texture, shape, and semantics, and accurately determine contextual relationships between pixels, while disregarding unnecessary parts of the feature map. Therefore, removing some of the unnecessary features and retaining the important ones is important to effectively improve the anti-interference ability of the model.

To reduce the impact of redundant information on the classification effect and to make full use of the contextual relationships between pixels, based on the threshold convolution [54] previously proposed by us, we propose a different REM from the threshold convolution, which consists of swish, tanh, and redundancy elimination operator.

The redundancy elimination operator with the DGA proposed in this article plays the core role of removing redundant information in HSIs in the REM. To better preserve rich semantic features, redundant features are reduced.

The swish activation function has a smooth curve shape that allows it to preserve the activation values of useful features while suppressing the influence of redundant features. The nonlinear characteristics of the Swish function help the model better distinguish differences between features and focus more attention on useful features. Therefore, the first branch of REM employs the Swish activation function to map each point on the feature map, aiming to minimize the impact of redundant features and improve the model's feature selection capability.

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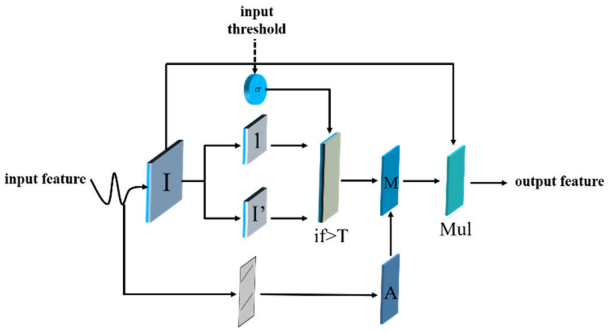


Fig. 3. Calculation process of the redundancy elimination operator. "I" represents the input feature map. "A" represents the values after the sigmoid activation function. "M" represents the process according to the threshold. "σ" represents the input threshold. "1" represents redundancy elimination, and "I'" represents the result of feature point conversion. Mul represents the dot product operation; the input and output features represent the feature map before the redundancy elimination convolution and after the redundancy elimination convolution, respectively.

The formula for calculation is given as follows:

$$m1_{ij}^{k} = \frac{m_{ij}^{k}}{1 + \exp(-m_{ij}^{k})} \tag{5}$$

where m_{ij}^{k} represents the feature points of the ith row and the jth column of the feature map m, and k is the index of the feature map.

The tanh activation function also has certain advantages in removing redundant features. The tanh function has an S-shaped curve, and its range is between -1 and 1. This nonlinear characteristic enables the tanh function to effectively map the values of features and push the activation values of redundant features toward the region close to 0. By compressing the activation values of redundant features into a smaller range, the tanh function helps reduce the impact of redundant features on the model and improves the model's focus on meaningful features. As a result, the second branch processes the convolution feature graph using the tanh function

$$m2_{ij}^{k} = \frac{e^{m_{ij}^{k}} - e^{-m_{ij}^{k}}}{e^{m_{ij}^{k}} + e^{-m_{ij}^{k}}}. \tag{6}$$

To bring the output features of the two branches into a reasonable range, the Hadamard product of the feature graphs $m1$ and $m2$ is employed for fusing the features of the output feature graph. The fusion of the above output feature graph can be expressed as follows: $(m1'^{*}m2') = m1_{ij}^{k} m2_{ij}^{k}$. $m1'$ and $m2'$ are the output characteristic maps of the first and second branches, respectively, and k represents the index of the characteristic maps.

Next, the output feature graph is deredundant (as shown in Fig. 3), and its core idea is to set $\sigma (0 < \sigma < 1)$, which represents the input threshold. For features lower than σ, it is assigned a value of 0 to eliminate this feature, while, for features higher than σ, it remains unchanged, which aims to remove more redundant information in HSIs and keep important feature information, thus reducing the amount of calculation, speeding up training convergence, and improving the accuracy.

The calculation formula is given as follows:

$$\mathrm{RE}(m_{ij}^{k}) = \begin{cases} (m_{ij}^{k})' = 0, & \text{if}(m1'^{*}m2') < \sigma \\ (m_{ij}^{k})' = m_{ij}^{k}, & \text{if}(m1'^{*}m2') \geq \sigma. \end{cases} \tag{7}$$

Among them, $0 \leq T \leq 1$ represents the threshold of the redundant operator (reoperator), $0 \leq \sigma \leq 1$; k is the index of the feature map; and m_{ij}^{k} represents the feature points in the ith row and the jth column of the feature map.

D. Phantom Fractal Module

Due to the excessive detail information of HSIs (a large part of which is redundant information) and the complexity of ground object features, the direct use of images for deep network feature extraction is easily disturbed by redundant information, resulting in incomplete network training and even training bias, which ultimately leads to poor feature description power and a weak image classification effect [55]. Especially when the amount of data is limited, single-scale neural networks struggle to extract useful information from the limited HSIs, leading to lower accuracy and slower convergence speed. The significant differences between the objects on the ground in HSIs are mainly in the texture and contour structure. To improve the feature description ability and integrate the feature information of objects of different scales, this article proposes a novel convolution to separate multiple parallel convolutions of different scales during the training process and inference process, which looks like "phantom operator" and is used for feature extraction of texture and contour structures to distinguish the feature in HSIs, so it is called PFM. The PFM takes the first scale as the main scale and, thus, fractals out multiple layers of convolution to form a multiscale feature map collection to process different ground features in a diversified form. The PFM will first divide the feature map into $s (s = 1, 2, 3, \ldots,)$ scales, followed by a direct separable convolution operation. A new feature map is then output, which is fused with the feature map of the first part for a 3*3 separable convolution of the $s (s = 2, 3, 4, \ldots)$ parts and so on. There is only one main scale for the module, and there are a total of $C_{s-1}^{0} + C_{s-1}^{1} + C_{s-1}^{2} + \cdots + C_{s-1}^{s-1}$, $s = 2, 3, 4, \ldots$, various combinations of feature maps for the sth scale. From this, the total number of receptive fields can be calculated, as shown in formula (8), with RF denoting the sum of the number of different receptive fields

$$\mathrm{RF} = \sum_{i=1}^{s} C_{s-1}^{0} + \sum_{i=2}^{s} C_{s-1}^{1} + \sum_{i=3}^{s} C_{s-1}^{2} + \cdots + C_{s-1}^{s-1}. \tag{8}$$

We demonstrate experimentally in Section IV-H3 that the best performance of the PFM can be achieved when $s = 6$, and therefore, the final fused feature maps will have one, two, four, eight, 16, and 32 different perceptual fields. Finally, a 1*1 convolutional layer is used to connect with the output of the PFM. The module works by enabling the model to extract more detailed feature textures, for instance, dense forest textures and flat road textures, greatly improving the accuracy of classification between different classes of features. The detailed structure of the PFM is shown in Fig. 4.

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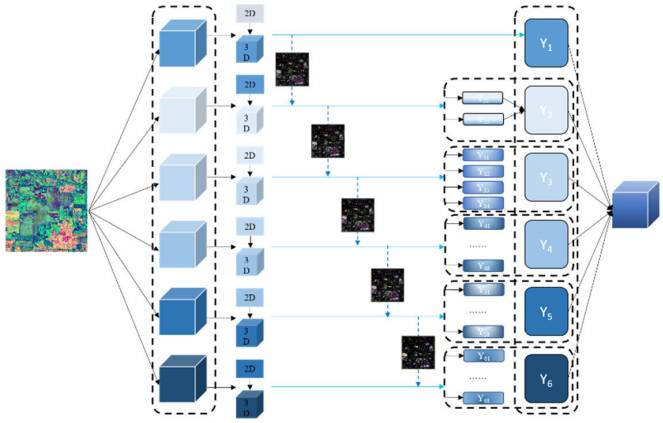


Fig. 4. Detailed structure of the PFM.

TABLE I
HARDWARE AND SOFTWARE PARAMETERS

Hardware environment		Software environment	
CPU	Intel(R) Xeon(R) Platinum 8255C CPU @ 2.50GHz	OS	Linux
RAM	40GB	CUDA	10.2
Video memory	11GB	Python	3.6.5
GPU	NVIDIA GeForce RTX 2080 Ti	Tensorflow	1.10.0
server platform	autodl	Keras	2.2.0

IV. EXPERIMENT

A. Experimental Environment and Settings

To eliminate differences in test results owing to the experimental environment and to assure the stability and repeatability of the program, all tests in this study are conducted using the same hardware and software. The specific environment version and parameters are shown in Table I.

We utilize RMSprop as the optimizer to make sure that the training is stable and to hasten model convergence. We set the batch size to 32, the initial learning rate to 0.001, the decay to 0.001, and the number of training rounds to 64 epochs taking into account the training time and the processing capability of the device. We set the run time to 10, which implies that, in order to acquire the data for all of the experimental findings in this study, we repeated the experiment ten times.

B. Data Description

To fully describe these datasets, we have provided the categories and sample sizes of the aforementioned datasets in Table II. Categories were condensed due to space constraints.

1) Indian Pines Dataset: The Indian Pines dataset was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor, which represents northwestern Indiana. It consists of 145*145 pixels with a spatial resolution of 20 m per pixel and 220 spectral wavelengths in the 0.4–2.5-μm band, with 16 classes of objects included in the dataset.

2) Salinas Dataset: It was gathered in California by the AVIRIS sensor and comprises 512*217 pixels with a spatial resolution of 3.7 m per pixel, 224 spectral wave bands, and 16 item categories.

3) DFC2018 Houston Dataset: The dataset for the 2018 IEEE GRSS Data Fusion Competition, called DFC2018 Houston, was created in Dr. Saurabh Prasad's Lab at the University of Houston. Multiple sensors were used to collect the data, which comprised 48 bands of hyperspectral data (1 m), three bands of light detection and ranging (LiDAR) data (0.5 m), and ultrahigh-resolution pictures (0.05 m) that covered 20 categories of ground items.

4) WHU-Hi: The WHU-Hi dataset, which was gathered and made available by Wuhan University's RSIDEA Research Group, can be utilized as the reference set for precise crop classification and HSI classification. The WHU-Hi dataset consists of three distinct unmanned aerial vehicle (UAV)-borne hyperspectral datasets: WHU-Hi-LongKou, WHU-Hi-HanChuan, and WHU-Hi-HongHu. All three datasets were collected in Hubei province, China, using a Headwall Nano-Hyperspec sensor mounted on a UAV platform in an agricultural area with a variety of crop types. WHU-Hi-HongHu has 22 categories, 270 bands, and 940 475 pixels. The intricacy of the data makes it possible to completely confirm the model's generalizability. As a result, WHUHi-HongHu is chosen to validate the model put forth in this study.

C. Evaluation Index

The average classification accuracy for each category is expressed in terms of the average accuracy (AA). It is calculated as follows:

$$\mathrm{AA} = \frac{1}{n}\sum_{i=1}^{n} \mathrm{recall}_i \tag{9}$$

where $recall_i$ denotes the classification accuracy of category i and n denotes the number of categories.

Overall accuracy (OA) represents the ratio between each correctly predicted sample and the total number of samples

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{FN}+\mathrm{FP}+\mathrm{TN}} \tag{10}$$

where TP represents the positive samples predicted by the model as the positive class, TN represents the negative samples predicted by the model as the negative class, FP represents the negative samples predicted by the model as the positive class, and FN represents the positive samples predicted by the model as the negative class.

Kappa is a comprehensive measure based on a confusion matrix and can be represented by k

$$k = \frac{N\sum_{i=1}^{r} x_{ii} - \sum_{i=1}^{r}(x_{i+}^{*}x_{+i})}{N^2 - \sum_{i=1}^{r}(x_{i+}^{*}x_{+i})} \tag{11}$$

where r denotes the total number of categories; x_{ii} denotes the number of correct classifications; x_{i+} and x_{+i} denote the total number of pixels in row i and column i, respectively; and N is the total number of pixels used for accuracy assessment.

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TABLE II
DATASETS' PARTITION FOR HSIS DATASETS

Indian Pines		Salinas		DFC2018 Houston		WHUHi-HongHu	
Class	**Samples**	**Class**	**Samples**	**Class**	**Samples**	**Class**	**Samples**
Alfalfa	46	Br-we-1	2009	Health-g	39196	Re-ro	14041
Corn-notill	1428	Br-we-2	3726	St-gr	130008	Road	3512
Corn-mintill	830	Fallow	1976	Ar-tu	2736	Ba-so	21821
Corn	237	Fa-pl	1394	Ev-tr	54322	Cotton	163258
Gr-pasture	483	Fa-sm	2678	De-trees	20172	Co-fi-wo	6218
Grass-trees	730	Stubble	3959	Bare-ea	18064	Rape	44557
Gr-pa-mo	28	Celery	3579	Water	1064	Ch-ca	24103
Hay-wi	478	Gr-un	11271	Re-bu	158995	Pakchoi	4054
Oats	20	So-vi-de	6203	No-re-bu	894769	Cabbage	10819
So-no	974	Co-se-we	3278	Roads	183283	Tu-mu	12394
So-mi	2455	Le-4wk	1068	Side-wa	136035	Br-pa-en	11015
So-cl	593	Le-5wk	1927	Cross-wa	6059	Br-ch	8954
Wheats	205	Le-6wk	916	Ma-th-fa	185438	Sm-br-ch	22507
Woods	1265	Le-7wk	1070	Hi-ways	39438	La-sa	7356
Bu-G-Dr	386	Vi-un	7268	Ra-ways	27748	Celtuce	1002
St-St-To	93	Vi-ve-tr	1807	Pa-pa-lo	45932	Fi-co-le	7262
				Un-pa-lo	587	Ro-le	3010
				Cars	26289	Carrot	3217
				Trains	21479	Wh-ra	8712
				St-seats	27296	Ga-sp	3486
						Broad-be	1328
						Tree	4040

D. Comparison of DGPF-RENet With State-of-the-Art Models

To further analyze the performance, we compared the proposed DGPF-RENet with nine representative HSI recognition methods using the python tool DeepHyperX, including traditional methods, CNN methods, and transformer methods, namely, SVM, 2-D-CNN [22], 3-D-CNN [23], RCNN [56], Yang [47], ViT [42], LeViT [43], RvT [45], and HiT [48].

SVM is a type of generalized linear classifier used for binary data categorization in supervised learning. It demonstrates numerous distinct advantages in solving limited sample, nonlinear, and high-dimensional pattern recognition problems. 2-D-CNN uses a CNN and AdaBoostSVM to learn the right representation for hyperspectral pictures, as well as for channel selection. 3-D-CNN is a 3-D HSI classification system that can process spectral and spatial data simultaneously. The advantages of DGPF-RENet in the integrated processing of spectral and spatial information will be compared to experimental results in Section IV-D. RCNN is a network designed for HSI classification using ResUnet residuals, which combines Unet's encoder–decoder backbone, residual connection, atrous convolution, pyramid scene parsing pool, and other features to reduce the model's data dependence with a small number of network layers. Yang is a finite element model that extracts effective spectral–spatial features from hyperspectral pictures using 3-D CNN and combined regularization. ViT is a typical transformer model. It has obtained specific results in the field of HSI classification using the benefit of self-attention. LeViT is a hybrid neural network for quick image categorization inference. In terms of speed/accuracy tradeoff, it outperforms previous Vits substantially. RvT improves the effect of self-attention by location-aware attention scaling and patch enhancement, which has good performance and strong robustness, by using robust components as the building blocks of ViT. HiT has included convolution operations in its transformer structure, which was specifically created for hyperspectral pictures, to capture small spectral changes and communicate local spatial context information. To extract tiny spatial–spectral distinctions, it employs a spectral adaptive 3-D convolution projection module and a convolution transposer. To create local spatial spectrum information from HSIs, the spectral adaptive 3-D convolution projection module employs two spectral adaptive 3-D convolution layers rather than linear projection layers. To reduce the possible errors in the experiment, we used the mean value of ten repeated trials to record the experimental values, eliminate errors as much as feasible, and keep two decimal points. In the categorization result chart, we display the result whose OA value ranks fifth out of ten experiments.

1) Results on the Indian Pines: The data collection on Indian pines has a tiny sample size and an extremely uneven distribution of 16 features. In contrast to the minimal sample size of 1% utilized by the Pavia University dataset and the Salinas dataset, a 5% training sample was chosen for the IP dataset in this study. This is because the categories "cut grass" and "oats" in the IP dataset contain only 28 and 20 samples, respectively. Even if 5% of the samples are chosen, only one can be used for training in this experiment, which is insufficient data for network training. In order to investigate the low data dependence of DGPF-RENet, 5% and 10% of the IP dataset were employed as the training set. The test results are shown in Table III.

Due to the tiny sample size of category 7 and category 9 data, there is only one training sample for 5% training data

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TABLE III
RESULTS ON THE INDIAN PINES DATASET

Train Sample	Class	Model									
		SVM	2D-CNN	3D-CNN	RCNN	Yang	ViT	LeViT	RvT	HiT	Ours
	1	0.00	0.00	21.61	**85.93**	62.78	38.17	0.00	10.41	77.73	62.95
	2	22.49	75.68	57.65	87.73	74.82	54.46	33.25	54.47	76.32	**91.06**
	3	0.00	66.40	40.00	80.46	57.79	44.13	1.51	40.70	58.07	**92.22**
	4	0.00	36.02	16.21	79.60	60.66	39.47	0.00	45.84	67.71	**91.68**
	5	0.02	70.27	70.45	86.16	67.39	30.06	10.95	28.06	73.48	**94.55**
	6	64.26	92.15	88.83	97.54	90.79	73.48	39.49	75.15	92.33	**95.89**
	7	0.00	0.00	9.36	**69.86**	37.57	35.63	0.00	0.00	47.60	60.37
	8	89.58	88.86	91.55	93.66	89.91	84.29	52.90	87.86	92.83	**99.63**
	9	0.00	0.00	6.67	61.32	51.79	31.14	0.00	8.00	53.27	**68.42**
5%	10	0.00	74.80	65.77	85.47	75.16	29.04	1.66	41.27	74.64	**92.63**
	11	55.59	86.80	72.71	91.66	78.98	60.97	71.18	53.28	81.65	**96.37**
	12	0.00	61.99	42.59	72.23	61.66	35.74	15.79	21.90	63.76	**91.56**
	13	0.00	96.92	89.26	**97.94**	91.83	84.08	15.52	72.06	88.54	96.46
	14	82.03	94.11	92.15	97.32	92.57	86.40	65.32	87.93	92.96	**98.94**
	15	0.00	49.15	18.75	59.92	53.12	41.84	8.28	20.49	52.15	**97.38**
	16	0.00	84.68	18.93	**90.67**	76.39	12.17	28.46	25.38	88.24	74.38
	OA	48.70	76.60	65.62	84.65	73.81	56.00	42.06	54.27	75.55	**94.44**
	AA	19.62	61.11	50.16	83.59	70.20	48.82	21.52	42.05	73.83	**87.78**
	k	37.19	73.37	60.46	82.62	70.29	49.77	32.92	48.01	72.24	**93.66**
	1	0.00	83.33	48.43	91.99	**94.50**	64.52	19.85	53.48	84.79	85.71
	2	37.12	93.52	67.88	**96.81**	89.38	66.10	56.52	72.94	84.71	96.40
	3	0.00	82.22	53.00	82.53	74.72	51.58	50.97	65.57	70.13	**96.47**
	4	0.00	86.64	41.50	82.15	78.79	58.58	34.19	82.12	77.95	**95.70**
	5	0.15	84.94	79.02	87.24	84.19	56.79	45.91	48.02	70.59	**96.28**
	6	66.97	98.07	92.48	**99.44**	97.73	81.90	82.82	82.99	95.00	97.63
	7	0.00	0.00	13.65	85.18	64.75	39.04	0.00	61.48	70.86	**87.69**
	8	88.33	93.56	92.90	94.29	94.11	88.64	84.39	91.65	92.32	**99.77**
	9	0.00	0.00	26.19	75.29	66.96	50.76	0.00	32.81	73.23	**90.0**
10%	10	0.00	87.66	71.35	92.69	85.56	58.23	39.52	70.53	83.51	**97.03**
	11	57.33	93.99	79.30	95.73	89.93	70.68	56.85	76.01	89.25	**98.10**
	12	0.00	88.08	58.67	91.69	82.04	53.22	56.35	61.78	78.22	**95.66**
	13	0.00	**99.91**	95.19	99.37	98.58	86.34	91.89	81.27	96.45	98.38
	14	82.79	97.20	94.11	97.33	96.35	89.90	92.44	90.13	92.76	**99.53**
	15	0.00	66.45	40.04	66.39	61.55	43.21	34.68	43.44	58.15	**98.30**
	16	38.67	93.11	45.03	**98.19**	94.20	39.11	87.27	88.83	88.22	87.74
	OA	50.80	87.25	72.37	89.15	84.43	66.02	59.51	71.66	81.46	**97.39**
	AA	23.21	78.04	62.42	89.77	84.58	62.41	52.10	68.94	81.63	**95.02**
	k	40.09	85.57	68.50	87.74	82.37	61.46	55.31	67.95	79.05	**97.02**
-	Params	-	489105	1543629	2738065	45822053	2607633	16946229	8938257	51231818	4228724

and two training samples for 10% training data under the condition of 5% training data. Normal neural networks require a specific quantity of training data, however, in order to detect appropriate feature information. Consequently, it poses significant difficulties for the feature learning of neural networks.

Table III displays the classification accuracy for each category of the Indian Pines dataset with 5% and 10% training sample sizes, as well as OA, AA, k, and trainable parameters. We examined the training outcomes for 5% Indian Pines. For category 7 and category 9 with only one sample, SVM and 2-D-CNN were incapable of recognition, 3-D-CNN, Yang, and other approaches yielded mediocre results, while RCNN and DGPF-RENet were the best learning models. Based on the above results, we note that the proposed DGPF-RENet model in this article has better classification performance than other deep learning-based methods. From the results of 5% training samples, the obtained OA is 94.44, AA is 87.78, and k is 93.66. In general, the index of OA, AA, and kappa increased by 9.79–45.74, 4.17–68.16, and 11.04%–56.47%, respectively.

Fig. 5 shows a plot of the visual classification results for the Indian Pines dataset, where we compare DGPF-RENet with nine other deep learning methods proposed in recent years for HSI classification, with the legend indicating the type of ground object represented by each color. Limited data collection is a defining characteristic of Indian Pines. DGPF-RENet only consumes 5% of the data compared to the other nine methods, but it has much less noise. It is closer to the truth and generally distinguishes different sorts.

2) Results on the Salinas: Compared to IP and PU datasets, the SA datasets' sample distribution is more evenly distributed, and the classification challenge is smaller. We used training sets of 0.1% and 1% for experimental verification in order to up the classification challenge and thoroughly evaluate the model's lack of data dependence. The test results are shown in Fig. 6.

Table IV shows the classification accuracy of the Salinas dataset for each category, as well as OA, AA, and k. Based on these results, we note that the model proposed in this article, DGPF-RENet, has better classification performance than other deep learning-based methods. On the Salinas dataset, the difficulty in classification is Vinyard_untrained and Grapes_untrained, i.e., class 8 and class 15. Other deep learning methods have difficulty achieving a good balance

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TABLE IV
RESULTS ON THE SALINAS DATASET

Train Sample	Class	Model									
		SVM	2D-CNN	3D-CNN	RCNN	Yang	ViT	LeViT	RvT	HiT	Ours
0.1%	1	0.00	27.92	21.35	39.63	8.69	0.00	0.00	0.00	0.45	**98.36**
	2	76.34	69.58	52.35	5.06	69.34	59.12	0.00	71.51	71.92	**99.86**
	3	0.00	14.09	37.78	68.33	40.73	0.00	0.00	44.23	65.40	**99.29**
	4	0.00	**87.66**	66.78	59.97	67.47	0.00	0.00	52.75	58.72	30.73
	5	0.39	77.40	47.83	79.90	73.41	0.00	0.00	28.95	71.80	**82.26**
	6	98.59	96.02	93.75	96.62	88.07	51.09	0.00	92.80	97.00	**99.52**
	7	86.01	67.00	53.77	17.38	72.09	0.00	0.00	64.90	66.10	**98.90**
	8	65.58	71.65	68.67	71.34	72.22	61.65	39.94	70.87	72.72	**90.46**
	9	54.16	81.76	74.62	89.66	83.37	52.94	0.00	83.04	84.37	**99.89**
	10	0.00	58.91	52.79	70.59	62.73	0.00	0.00	53.75	66.37	**91.37**
	11	0.00	0.00	10.91	32.84	42.40	0.00	0.00	0.48	8.97	**86.69**
	12	0.00	52.01	48.77	83.67	51.73	0.00	0.00	63.69	67.97	**88.21**
	13	0.00	31.75	67.70	44.22	56.61	0.00	0.00	29.41	63.60	**77.30**
	14	0.00	0.80	45.08	60.78	64.04	0.00	0.00	**73.36**	71.67	68.35
	15	0.00	6.61	16.43	29.49	35.51	3.19	0.00	2.37	52.92	**54.16**
	16	0.00	6.44	1.78	30.85	18.07	0.00	0.00	10.60	30.04	**97.21**
	OA	52.24	62.54	57.24	60.82	64.30	41.20	20.15	60.68	67.50	**86.67**
	AA	23.82	46.85	47.52	55.02	56.66	14.25	2.50	46.42	59.38	**85.16**
	k	44.38	57.61	51.75	56.43	60.10	31.71	0.58	55.60	63.67	**85.07**
1%	1	11.87	93.94	92.15	94.31	83.89	27.70	70.14	56.58	85.09	**99.74**
	2	78.63	99.80	98.79	99.82	94.08	80.78	87.75	82.30	95.65	**99.99**
	3	0.00	95.60	94.26	98.44	91.40	95.02	35.25	89.46	94.33	**99.95**
	4	96.37	97.52	96.04	97.83	96.21	92.62	81.51	88.30	95.27	**98.86**
	5	72.69	97.56	96.61	98.57	95.23	93.25	77.63	90.16	97.18	**99.40**
	6	99.02	97.57	97.14	97.64	97.32	97.38	94.50	96.91	97.58	**99.85**
	7	90.12	97.15	96.53	97.99	95.89	92.22	92.71	95.64	95.73	**99.71**
	8	67.95	89.69	80.40	95.19	83.72	80.09	81.02	80.49	87.14	**95.72**
	9	68.97	98.10	97.14	99.25	96.59	98.14	90.23	96.58	97.97	**100.00**
	10	10.35	93.49	91.59	95.09	90.88	89.51	76.33	82.24	93.64	**99.47**
	11	0.00	84.11	87.72	95.23	83.45	80.74	17.81	64.93	88.01	**99.31**
	12	0.12	96.55	96.04	96.56	93.52	90.65	72.58	88.93	94.34	**99.11**
	13	0.00	97.01	93.22	96.99	95.29	89.97	88.55	87.53	94.54	**98.25**
	14	72.68	96.47	94.57	96.08	93.74	95.26	52.49	91.80	96.11	**97.04**
	15	0.00	80.82	61.93	90.33	69.83	69.32	59.79	71.76	74.48	**97.00**
	16	17.25	77.44	73.49	79.97	75.39	54.00	43.30	67.97	72.56	**99.36**
	OA	61.88	90.10	85.23	93.09	85.80	81.86	75.36	81.77	87.92	**98.42**
	AA	42.88	93.30	90.48	95.58	89.78	82.92	70.10	83.22	91.23	**98.92**
	k	56.17	89.03	83.61	92.35	84.26	79.90	72.69	79.84	86.60	**98.24**
-	Params	-	491409	1604829	2740369	46617639	2612241	16948277	8942865	51656456	4277812

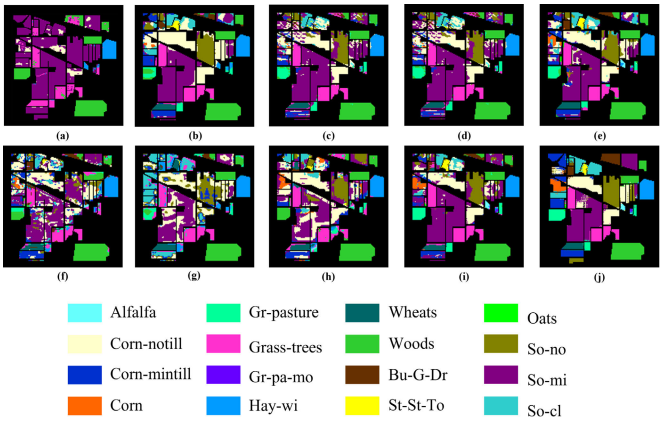


Fig. 5. Classification visible results for the Indian Pines HSI by (a) SVM, (b) 2-D-CNN, (c) 3-D-CNN, (d) RCNN, (e) Yang, (f) ViT, (g) LeViT, (h) RvT, (i) HiT, and (j) Ours.

between these two datasets, which can easily lead to misclassification. The DGPF-RENet proposed in this article, on the other hand, balances the two well and achieves relatively good classification results in both categories, probably because REM is good at dealing with the contextual relationships between pixels. From the results of 0.1%, the obtained OA was 86.67%, AA was 85.16%, and kappa was 85.07%.

Fig. 6 shows a graph of the visual classification results for the Salinas dataset, where we compare DGPF-RENet with six other deep learning methods proposed in recent years for HSI classification, with the legend indicating the type of ground object represented by each color. In the Salinas dataset, Vinyard_untrained and Grapes_untrained are very difficult to distinguish from each other because of their spatially similar texture and contour features. Other methods struggle to meet this challenge due to insufficient feature learning capabilities. In contrast, our proposed DGPF-RENet has a higher discrimination power. We have analyzed the reasons for the improved classification accuracy, which mainly lie in the processing of redundant information by REM and the powerful feature learning capability of PFM.

3) Results on the DFC2018·Houston: The 2018 Data Fusion Contest, run by the IEEE Geoscience and Remote Sensing Society Image Analysis and Data Fusion Technical Committee, produced the scientifically significant DFC2018Houston dataset. The 2018 competition focuses on

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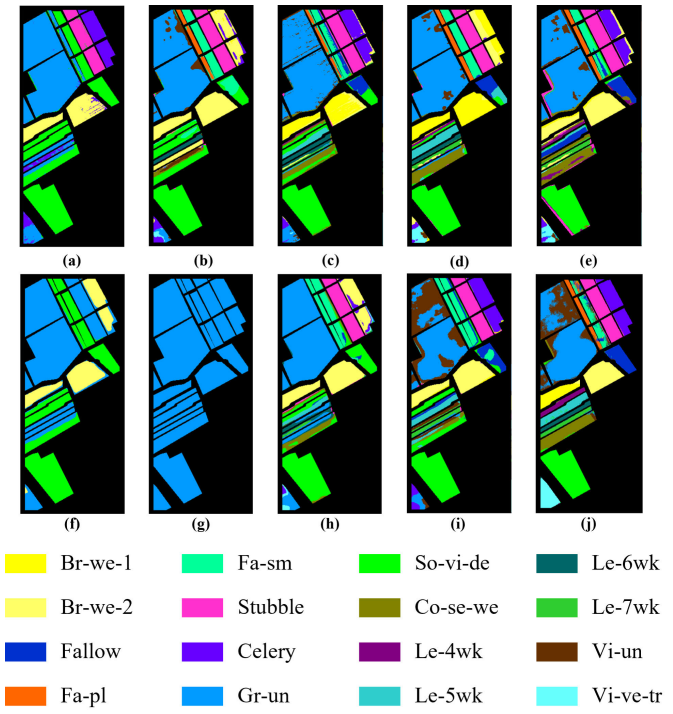


Fig. 6. Classification visible results for the Salinas HSI by (a) SVM, (b) 2-D-CNN, (c) 3-D-CNN, (d) RCNN, (e) Yang, (f) ViT, (g) LeViT, (h) RvT, (i) HiT, and (j) Ours.

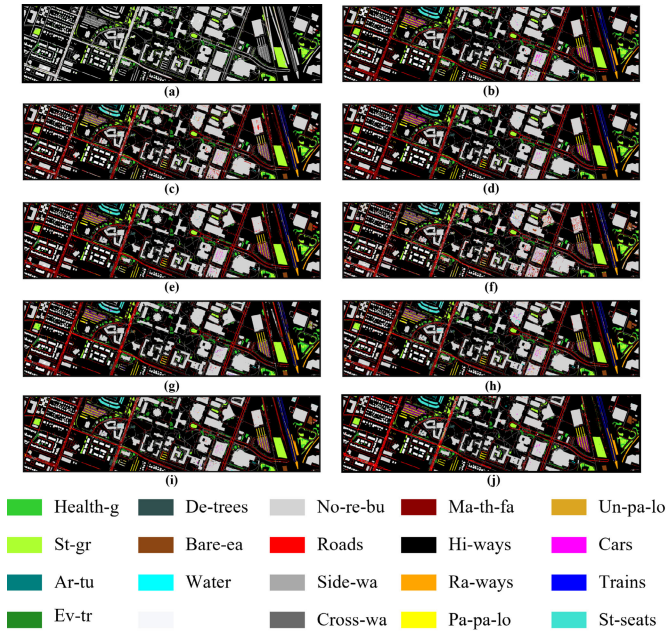


Fig. 7. Classification visible results for the DFC2018 Houston HSI by (a) SVM, (b) 2-D-CNN, (c) 3-D-CNN, (d) RCNN, (e) Yang, (f) ViT, (g) LeViT, (h) RvT, (i) HiT, and (j) Ours.

sophisticated multisource optical remote sensing for urban observation and monitoring (multispectral LiDAR, hyperspectral imaging, and high-resolution imagery). The competition tries to identify between the incredibly diverse and intricate categories of urban items, materials, and vegetation based on the classification of urban land use and land cover. Table V displays the test results.

The DFC2018 Houston data collection has a limited number of bands and large spatial coverage. Compared to other hyperspectral datasets, this one contains a small number of bands. Moreover, the DFC2018 Houston dataset is distinguished by an imbalanced sample size. Due to the fact that the 0.2% training set is immediately subdivided for training, the category with the largest total sample size has a greater number of training samples. The category with the greatest number of samples has a significant impact on the OA index, which, in turn, impacts the presentation of the benefits of DGPF-RENet. Nevertheless, the effect of categories with small sample sizes on the overall classification cannot be ignored. We should simultaneously pay attention to OA and AA signs. In general, DGPF-RENet calculated the tradeoff between OA and AA indices, taking into account the classification accuracy of different categories. Despite being marginally inferior to RCNN in the OA index, the AA index obtained clear advantages and outperformed the other eight methods in the OA, AA, and kappa indices.

The categorization outcomes of the ten approaches are displayed in Fig. 7. DGPF-RENet, in contrast, has less noise and a more evenly distributed categorization result.

4) Results on the WHUHi-HongHu: The experimental region where multiple cultivars of the same crop are cultivated is the source of the WHUHi-HongHu data collection. There are several species, which makes classification challenging. The test results are shown in Table VI.

The WHUHI-HongHu dataset is on a vast scale. Some Whuhi-Honghu dataset categories cover tens of thousands or even hundreds of thousands of sample points, and there is a class difference. In the experiment, we only employ 25 or 50 sample points for training each class, which poses significant obstacles to the model's feature learning. DGPF-RENet outperformed other models in terms of AA index, indicating DGA's superiority in learning data features under conditions of minimal data sample reliance.

Fig. 8 shows a plot of the visual classification results for the WHUHi-HongHu dataset, where we compare DGPF-RENet with nine other deep learning methods proposed in recent years for HSI classification, with the legend indicating the type of ground object represented by each color. This dataset is substantial in scope. DGPF-RENet employs fewer sample points in each category than the other nine methods—only 25 per category. Even if the cotton group has more noise, overall, all of the categories have been roughly distinguished and are closer to the ground truth.

E. Comparison of Representative Few-Shot Methods

In this section, to fully validate the superiority of DGPF-RENet in the aspect of small-sample learning, we selected currently novel small-sample methods for comparison and applied the same data partitioning to the training samples. The specific partitioning ratios are given as follows: Indian Pines (5%), Salinas (0.1%), DFC2018 Houston (0.2%), and WHUHi-HongHu (25 per class). The experimental results can be seen in Table VII. The small-sample methods used for comparison include A-SPN [57], BTANet [58],

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TABLE V
RESULTS ON THE DFC2018 HOUSTON DATASET

Train Sample	Class	Model									
		SVM	2D-CNN	3D-CNN	RCNN	Yang	ViT	LeViT	RvT	HiT	Ours
	1	75.56	85.72	85.43	86.42	87.81	83.48	72.49	86.97	**87.89**	80.10
	2	79.08	92.59	92.25	**93.15**	92.89	89.25	84.64	91.07	92.09	93.08
	3	0.00	85.11	76.50	84.87	75.21	29.12	0.00	66.05	72.81	**99.74**
	4	83.34	93.93	93.30	94.01	92.61	90.84	87.76	90.95	93.29	**97.91**
	5	0.00	70.08	69.31	78.91	71.15	49.70	36.76	66.40	73.95	**86.78**
	6	0.00	92.21	75.33	93.24	84.24	69.18	50.13	90.40	89.33	**98.53**
	7	9.48	70.01	12.72	84.27	59.24	2.95	49.25	39.15	68.84	**100.00**
	8	45.50	79.92	76.38	84.45	78.17	70.07	76.66	78.27	79.60	**93.63**
	9	69.59	94.28	90.01	95.16	93.07	83.09	93.13	91.93	92.89	**97.55**
	10	0.00	60.17	55.69	71.68	63.19	49.57	53.38	62.41	64.58	**82.09**
	11	27.13	64.29	55.99	68.86	60.92	47.95	54.51	57.97	60.69	**71.57**
0.2%	12	0.00	0.00	2.14	16.30	1.69	3.60	0.00	9.82	5.26	**28.09**
	13	8.11	72.70	62.70	**79.16**	68.09	55.81	58.59	71.88	71.96	29.43
	14	0.00	78.50	64.13	85.79	76.16	69.36	54.57	79.44	82.96	**93.39**
	15	0.00	91.42	91.18	96.55	91.92	76.26	81.85	91.59	89.75	**99.58**
	16	0.00	82.63	70.24	91.36	80.48	53.13	80.49	74.04	79.37	**94.46**
	17	0.00	0.00	2.57	62.73	3.32	2.12	0.00	13.70	2.99	**100.00**
	18	0.00	77.10	51.43	81.30	61.75	30.77	55.57	54.04	56.77	**88.20**
	19	0.00	89.80	80.25	**94.32**	85.89	71.36	60.99	88.97	88.21	92.97
	20	0.00	91.07	81.27	92.61	91.37	64.64	79.39	80.06	82.82	**99.29**
	OA	56.58	84.09	78.70	**87.39**	82.44	71.81	77.73	81.76	83.04	86.63
	AA	19.89	73.58	64.44	81.76	70.96	54.61	56.51	69.26	71.80	**86.32**
	k	33.34	79.45	72.16	83.59	77.24	63.13	71.04	76.19	77.94	**82.63**
-	Params	-	402581	467233	2659733	17621261	2433045	16870457	8763669	41073050	2222136

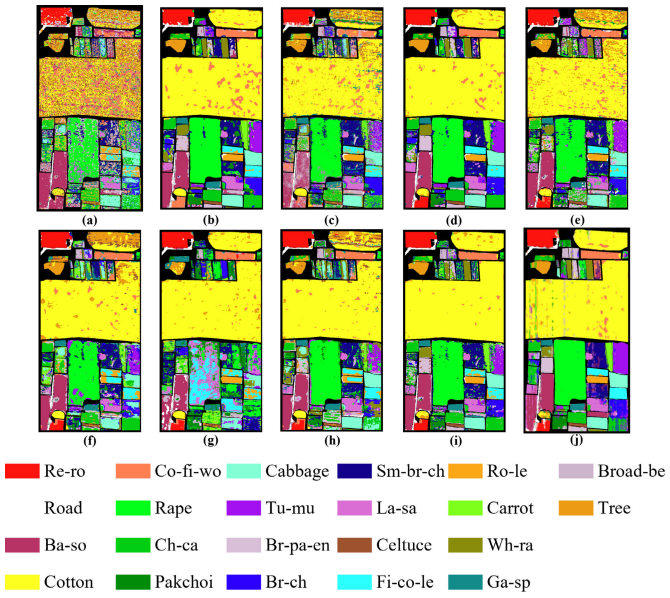


Fig. 8. Classification visible results for the WHUHI-HongHu HSI by (a) SVM, (b) 2-D-CNN, (c) 3-D-CNN, (d) RCNN, (e) Yang, (f) ViT, (g) LeViT, (h) RvT, (i) HiT, and (j) Ours.

DBDA [59], and CSIL [60]. A-SPN is an attention-based second-order pooling network. The proposed pooling operator generates more discriminative and compact feature representations by utilizing second-order statistics and data-adaptive weights, thereby significantly improving the utilization efficiency of pixel information by the model. BTANet focuses on eliminating redundant information and mining promising features. It introduces a neural network with concatenated three-way attention mechanisms and bole convolution for HSI classification. DBDA proposes a dual-branch dual-attention mechanism network for HSI classification. The two branches are designed to capture a large number of spectral and spatial features contained in HSI, and refine and optimize the extracted feature maps, thus achieving the effect of reducing the training samples. CSIL is an interactive learning framework inspired by ViT from a central to surrounding perspective. It achieves multiscale, detail-aware, and spatial interactive classification based on carefully designed hierarchical region sampling strategies, central transformers, and surrounding transformers, achieving the effect of small-sample training.

The experimental results show that our proposed DGPF-RENet achieves higher accuracy under the extremely limited training sample conditions set in this article compared to recent representative few-shot methods. This demonstrates that DGPF-RENet outperforms these recent representative methods. The superiority of DGPF-RENet on four datasets highlights its generalization and versatility.

F. Computational Complexity

In order to provide a detailed comparison of the computational complexity between DGPF-RENet and other models, important metrics, such as floating point operations (FLOPs), training time, and testing time, are presented in Table VIII.

It is worth noting that, for the sake of convenience and fair comparison of computational complexity, the input image size is set as 1*200*15*15. From this, we can observe that the efficiency of DGPF-RENet is not optimal, but it surpasses most of the compared methods, achieving a balance between model efficiency and accuracy under small sample conditions. The proposed method in this article primarily focuses on improving accuracy and accelerating training convergence under small sample conditions, while it falls short in terms of

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TABLE VI
RESULTS ON THE WHUHI-HONGHU DATASET

Train Sample	Class	Model									
		SVM	2D-CNN	3D-CNN	RCNN	Yang	ViT	LeViT	RvT	HiT	Ours
25 per class	1	0.00	82.52	76.48	91.12	86.40	84.85	81.11	83.12	**92.62**	87.27
	2	71.58	76.05	70.11	81.79	78.17	73.05	68.50	65.81	86.74	**87.89**
	3	48.31	85.39	76.20	87.68	83.43	81.47	72.23	81.42	84.98	**87.62**
	4	65.21	93.82	89.37	96.25	88.42	90.29	93.42	93.21	**97.01**	92.30
	5	62.57	47.45	45.84	74.83	50.48	58.11	71.69	68.78	83.79	**94.17**
	6	7.89	92.34	87.66	93.80	90.23	87.04	47.97	87.13	91.57	**94.77**
	7	67.22	73.28	51.66	**77.82**	61.56	51.51	34.54	46.27	75.94	64.71
	8	41.13	38.19	23.62	53.38	28.12	23.16	22.53	24.60	45.74	**79.23**
	9	2.95	94.97	81.06	**95.55**	90.73	53.67	85.75	71.96	95.05	96.13
	10	64.35	72.39	54.97	**80.11**	74.05	52.64	42.54	56.94	64.24	77.71
	11	9.88	67.89	28.44	71.26	43.61	31.38	11.14	39.38	58.95	**85.90**
	12	12.20	58.64	52.62	**73.96**	64.03	43.81	20.37	47.91	63.46	70.50
	13	34.14	68.54	63.38	**73.80**	66.57	59.47	35.53	60.50	67.75	68.81
	14	36.45	84.77	60.93	**89.47**	66.59	67.14	44.89	69.46	83.71	88.11
	15	44.18	83.28	26.48	88.64	36.10	15.56	45.42	22.34	76.86	**98.98**
	16	3.94	91.52	85.73	95.27	90.99	61.64	65.05	76.23	82.84	**94.11**
	17	58.49	81.51	76.30	89.54	86.20	30.65	71.04	42.70	71.07	**98.82**
	18	34.09	60.00	39.33	81.80	52.19	34.30	18.69	48.42	70.35	**97.50**
	19	10.86	81.52	52.26	83.28	69.24	36.31	29.76	42.71	76.52	**87.75**
	20	6.46	78.04	56.48	80.61	66.50	53.62	40.82	48.39	77.50	**92.85**
	21	9.45	34.12	25.05	52.42	35.24	35.08	20.63	34.54	43.07	**99.85**
	22	6.95	65.57	35.57	67.56	29.17	38.01	29.95	50.73	72.24	**95.23**
	OA	46.13	81.98	71.63	87.08	75.59	70.40	64.25	74.27	84.85	**87.88**
	AA	31.74	73.26	57.25	80.91	65.36	52.85	47.89	57.39	75.55	**88.19**
	k	39.27	77.85	65.70	83.95	70.60	64.12	56.34	68.47	81.13	**84.94**
50 per class	1	77.02	87.91	88.01	92.11	88.83	88.24	88.75	87.50	91.85	**95.81**
	2	54.75	82.84	83.80	86.10	84.30	75.08	71.98	75.41	85.78	**93.01**
	3	78.60	86.47	86.19	88.52	85.11	85.97	77.89	86.15	87.30	**89.18**
	4	65.54	94.47	93.40	**97.18**	90.95	91.49	93.47	94.01	96.64	95.77
	5	13.56	50.18	67.13	87.54	60.52	67.05	67.96	70.43	81.47	**96.70**
	6	73.48	92.35	90.36	93.95	93.47	87.04	86.62	89.14	92.13	**97.21**
	7	48.24	77.94	60.14	77.83	66.83	54.37	55.57	59.94	75.38	**79.24**
	8	4.50	51.67	36.79	58.69	43.17	25.93	30.49	36.71	44.49	**94.39**
	9	77.84	95.26	92.47	**96.42**	93.48	74.64	93.44	77.87	94.96	96.20
	10	21.99	77.05	69.98	81.33	76.05	59.13	63.48	66.26	75.84	**88.70**
	11	6.53	71.10	47.03	74.52	55.68	38.83	45.71	42.81	66.40	**89.90**
	12	41.69	67.75	63.98	68.01	69.18	52.41	46.35	60.80	70.12	**77.04**
	13	28.45	71.32	69.08	71.94	74.00	59.26	40.01	66.21	71.31	**78.81**
	14	49.86	89.76	76.46	90.96	71.98	66.67	74.68	76.46	83.31	**92.44**
	15	6.39	89.32	37.98	95.56	57.76	28.51	39.62	28.08	82.83	**98.70**
	16	71.10	92.42	87.32	94.21	94.14	70.53	91.49	78.82	88.46	**97.16**
	17	26.24	82.60	79.34	87.83	91.29	51.71	82.20	63.74	79.86	**98.32**
	18	10.54	71.88	59.67	76.33	56.05	40.99	35.43	44.01	65.20	**99.25**
	19	14.83	84.18	72.54	86.56	77.27	45.10	74.39	49.65	76.44	**93.43**
	20	13.82	86.00	72.16	83.28	75.77	53.62	46.13	57.02	79.78	**99.06**
	21	10.64	39.09	43.37	55.20	41.47	44.54	29.39	40.02	49.00	**99.97**
	22	19.06	77.66	46.57	71.05	34.85	36.30	39.25	50.14	66.50	**99.01**
	OA	49.85	84.60	80.32	87.93	80.34	74.08	75.98	77.96	85.68	**92.75**
	AA	37.03	78.15	69.26	82.51	71.92	58.97	62.47	63.69	77.50	**93.15**
	k	43.00	81.00	75.80	84.95	76.11	68.43	70.54	72.94	82.19	**90.91**
-	Params	-	530967	2828835	2792215	59851056	2689047	16985147	9019671	60112172	5149498

testing time, which presents certain limitations in the practical applications of the model. In terms of model efficiency, future efforts can be directed toward utilizing model compression techniques to further reduce the size of the model and speed up training and testing.

G. Effects of Different Numbers of Training Samples

To further validate the low data dependence characteristics of DGPF-RENet and investigate the impact of training sample quantity on its performance, we conducted additional experiments with different training sample partitions. The aim was to achieve a more comprehensive analysis. We divide the Indian Pines dataset into {5%, 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 99%} training samples, Salinas into {0.1%, 1%, 5%, 10%, 20%, 40%, 50%, 60%, 70%, 80%, 99%}, and DFC2018 Houston into {0.2%, 1%, 5%, 10%, 20%, 40%, 50%, 60%, 70%, 80%, 99%}.

Fig. 9 shows the variations of OA, AA, and kappa under different training sample ratio settings in order to explore the impact of limited data on the performance of DGPF-RENet. Generally, increasing the number of training samples is beneficial for feature extraction and obtaining better performance

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TABLE VII
COMPARISON OF THE LATEST FEW-SHOT METHODS

Dataset		A-SPN	BTANet	DBDA	CSIL	Ours
Indian Pines	OA	**94.68**	93.56	93.31	93.77	94.44
	AA	83.78	70.62	84.31	86.88	**87.78**
	k	93.42	92.64	91.20	92.90	**93.66**
Salinas	OA	86.46	**87.27**	86.54	84.14	86.67
	AA	79.18	83.95	81.34	77.79	**85.16**
	k	84.78	**85.81**	83.64	82.36	85.07
DFC2018 Houston	OA	84.86	**87.04**	84.55	85.44	86.63
	AA	75.20	74.46	83.14	79.84	**86.32**
	k	80.61	81.04	**84.27**	82.25	82.63
WHUHi-HongHu	OA	86.79	85.57	86.31	87.32	**87.88**
	AA	87.23	86.69	84.55	84.12	**88.19**
	k	83.63	82.18	**86.24**	86.11	84.94

TABLE VIII
COMPUTATIONAL COMPLEXITY OF MODELS, INCLUDING FLOPS, TRAINING TIME, AND TESTING TIME

Methods	Flops(GB)	Training time(s)	Testing time(s)
2DCNN	0.07	15.91	1.20
3DCNN	0.27	88.02	3.54
RCNN	3.88	31.61	1.92
Yang	7.62	99.12	4.68
ViT	0.14	62.89	2.08
LeViT	1.81	148.51	7.07
RvT	0.42	67.55	3.68
HiT	2.33	112.08	6.71
A-SPN	0.12	8.34	1.93
BTANet	0.14	13.34	3.76
DBDA	0.06	68.69	4.17
CSIL	0.74	306.27	164.58
Ours	0.26	32.06	4.96

from the model. For the Indian Pines dataset, some classes have a small number of total samples, and under the 5% training data split, these classes have only one or a few training samples. This makes it difficult for the model to obtain effective features for these classes, thus limiting the model's performance. However, DGPF-RENet still performs well with a 5% data split. When the training samples increase to 10%, the model's performance significantly improves, and the performance reaches saturation at a training sample ratio of around 40%. For the Salinas dataset, the total number of samples is relatively larger compared to the Indian Pines dataset. Satisfactory model performance is achieved even with a very low number of training samples, and the performance reaches saturation at around 10% of the data. For the Houston 2018 dataset, its high resolution and large-scale characteristics limit the model's performance. It can be observed that there is a significant difference in the model's performance under the settings of 0.2% and 1% training samples. There is a high degree of imbalance among different classes, which restricts the model's performance in terms of AA metric.

H. Experiments for the Determination of Key Module Parameters

In this section, we determine the optimal parameters for each of the modules we designed and discuss their final

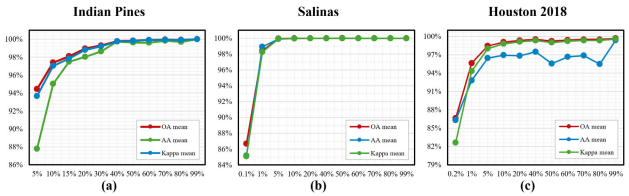


Fig. 9. Variation of OA, AA, and kappa with the number of training samples. (a) IP. (b) PU. (c) DFC2018 Houston.

results in detail. We keep the other factors of DGPF-RENet unchanged and only modify the parameters of a single module to demonstrate the applicability of the parameters set in this article for HSIs. The appropriate parameters for the DGA weight-matching strategy are discussed in Section IV-H1. The thresholds for REM are determined in Section IV-H2. The results for setting different scales of the PFM are compared in Section IV-H3.

1) DGA Weight-Matching Strategy Parameter Determination: The values of α, β, and γ will influence the effectiveness of the DGA, as it relates to the weight-matching strategy for c_1, c_2, and c_3. As the attention mechanism focuses on the key feature information in the image, this experiment is based on the premise

$$\alpha > \beta \geq \gamma. \tag{12}$$

We have experimented with different values for α, β, and γ. The results are shown in Fig. 10.

Experiments show that the best values are $\alpha = 0.7$, $\beta = 0.2$, and $\gamma = 0.1$. The reason is that, when α is too large, the DGA loses the global nature of the image features. In contrast, attention is not focused enough on important feature information, thus impacting the effectiveness of DGA.

2) Threshold Determination of the REM: For redundancy elimination convolution in the REM, threshold setting is an important step. We set the threshold σ for five different values and designed experiments with values of 0.1, 0.15, 0.2, 0.25, and 0.3. The results of the experiment are shown in Fig. 11.

Experiments have shown that the optimum threshold σ is 0.2. The reason for σ being too large is that too many feature points are eliminated, so the model performs poorly at values of 0.25 and 0.3. In contrast, when σ is too small, the elimination of feature points is not sufficient, and more redundant information will still remain in the feature map.

Nevertheless, we found that the experimental results were not significantly affected when the threshold was varied between 0.1 and 0.3, so that good results could be achieved without overly precise threshold adjustment, demonstrating that the REM is reasonably well adapted to the threshold.

3) Determination of the Number of Scales for the PFM: To demonstrate the optimal scale on which the PFM should be set, we set ten different values for s from 1 to 10. AA for the three datasets with different values of s is shown in Fig. 12.

The model works best when the scale s is taken to be 6. We believe that the reason for this is that, when the value of s is too small, the PFM does not have enough feature learning ability, and when the value of s is too large, the model tends to overfit and, thus, does not perform well on the test set.

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TABLE IX
ABLATION EXPERIMENT RESULTS

Group	Indian Pines			Salinas			DFC2018 Houston			WHUHi-HongHu		
	OA	AA	k	OA	AA	k	OA	AA	k	OA	AA	k
1	94.44	87.78	93.66	86.67	85.16	85.07	86.63	86.32	82.63	87.88	88.19	84.94
2	90.26	83.35	89.29	80.06	81.88	81.91	86.15	82.13	81.56	85.12	85.68	83.16
3	91.35	84.70	89.83	82.19	83.74	83.56	84.35	84.11	80.64	86.68	87.23	83.99
4	90.79	83.18	89.15	82.65	83.38	83.60	84.28	84.08	79.96	85.61	85.94	83.46
5	83.51	76.32	83.19	80.01	81.14	81.86	81.85	81.33	78.91	84.11	84.25	82.47
6	78.84	75.72	78.24	75.12	77.24	77.61	79.77	79.68	76.97	80.00	80.33	77.88
7	83.39	79.34	83.97	80.28	80.19	80.72	80.04	76.37	75.51	84.61	84.95	82.13
8	81.16	78.11	80.58	77.35	79.27	79.54	78.11	74.34	73.25	82.34	82.69	79.11

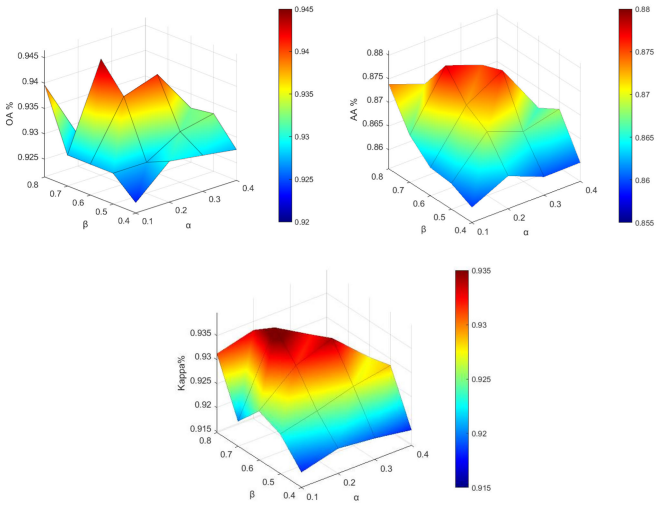


Fig. 10. Relationships between OA, AA, kappa, and α, β, respectively.

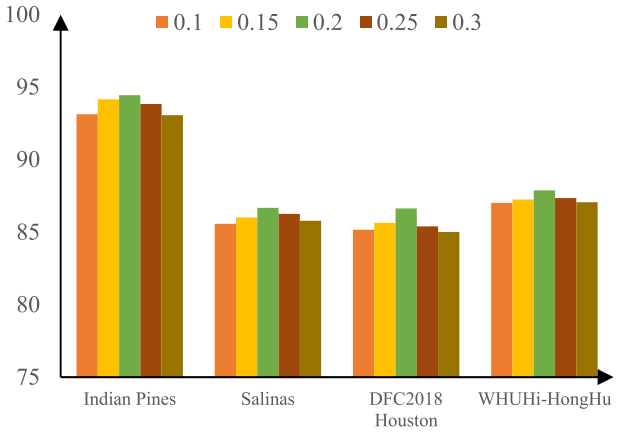


Fig. 11. Relationship between OA and threshold setting in four datasets; different colors represent different REM threshold settings.

I. Ablation Experiment

To verify the effectiveness of the method proposed in this article, we implemented different variants of DGPF-RENet. In addition, we display the ablation experiment that divided the training sets into the fewest numbers in each of the four datasets since this can more accurately depict the minimal data dependence of DGPF-RENet. In each implementation, we delete DGA using the controlled variables method, REM with hybrid dilation convolution in REM, and PFM with ordinary convolution (for ease of presentation, the term "remove" is used in this section to denote the replacement of the original DGPF-RENet modules with the corresponding conventional methods). Among them, Group 1 represents DGPF-RENet, Group 2 removes DGPF-RENet's DGA, Group 3 removes DGPF-RENet's REM, Group 4 removes DGPF-RENet's PFM, Group 5 removes DGPF-RENet's DGA and REM, and Group 6 removes DGPF-RENet's DGA and PFM. Group 7 removed REM and PFM from DGPF-RENet, and Group 8 removed all three modules of DGA, REM, and PFM from DGPF-RENet. The experimental results are given in Table IX.

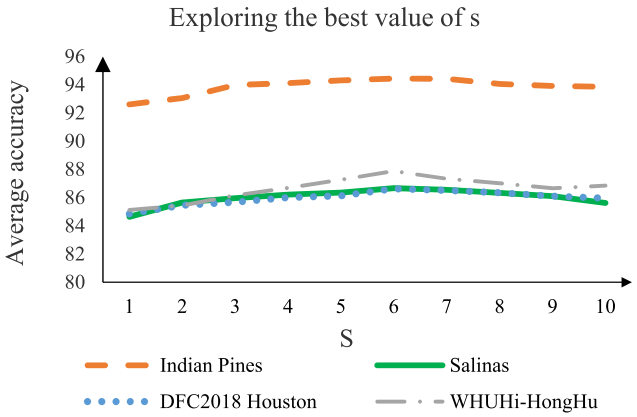


Fig. 12. Exploring the experimental results of the value of s.

We can make the following observations.

1) The comparison of Groups 1 and 5–8 shows that PFM has the most significant improvement when these methods are applied singularly, while REM leads to a decrease in accuracy. The reason for this is that PFM improves the learning ability of the model based on the experimental conditions in this article having fewer training samples and fewer iterations, in which case the improvement in learning ability directly improves the final results. This also demonstrates the portability of PFM. REM, on the other hand, brings about a decrease in accuracy due to the removal of redundant features and the loss of feature information when applying REM singularly. Therefore, the portability of REM is poor and difficult to apply directly to other networks.

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2) The comparison of Groups 4, 6, and 7 shows that using DGA and REM together can result in a relatively large improvement in the results. It is worth noting that Group 4 eliminates the problem of the model accuracy decreasing using REM alone. This demonstrates the importance of assigning appropriate weights to spatial–spectral information while removing redundant information.
3) The comparison of Groups 1–4 showed that Group 4 exhibited the best results compared to Groups 2 and 3, demonstrating the superiority of the combined use of DGA and REM due to the removal of redundant information while assigning appropriate weights to the spatial–spectral information. Group 1 finally showed the best experimental results, which demonstrates the superiority and necessity of the three methods proposed by DGPF-RENet.
4) The contributions of DGA, REM, and PFM to the accuracy improvement and their necessity can be well demonstrated by the comparative results of eight sets of experiments; therefore, this article concludes that DGPF-RENet is a highly accurate method based on CNN architecture suitable for HSI classification.
5) In comparison to the other two modules, DGA has the biggest influence on the AA index for the DFC2018 Houston dataset because it enables the model to concentrate on comprehending and learning categories with fewer training samples, leading to an improvement in these categories.

V. Discussion

The data reliance and training convergence rounds of the model are decreased by DGPF-RENet compared to the current state-of-the-art approach. DGPF-RENet is defined by the use of visual glances to concentrate on significant 3-D characteristics and remove as much unnecessary information from hyperspectral pictures as feasible. The benefits of DGPF-RENet in terms of low data reliance and quick convergence are then briefly discussed, along with any potential drawbacks to this work.

A. Model Principle Explanation

This article presents a novel network architecture for high-dimensional HSI classification. The network introduces three new modules: DGA, REM, and PFM, with the aim of reducing the model's data dependence and accelerating the training convergence process. This section provides a detailed explanation of the principles behind these three modules and their effectiveness in low-data dependence scenarios.

1) DGA: Inspired by the human visual system's ability to perceive subtle differences between objects, including texture and shape, this study observes the research on GG-transformer and discovers that self-attention mechanisms and convolutions can simulate the global perceptual ability of the human eye. This observation serves as the inspiration for the development of DGA. DGA utilizes the concept of "glancing" to simulate the selection process of the human eye's gaze points, allocating appropriate weights to the spectral and spatial dimensions. It achieves the extraction of horizontal and vertical spatial continuity, feature similarity between adjacent spectral bands, and differentiation of different categories in continuous positions. This feature extraction method effectively utilizes the feature information from limited samples and reduces the model's dependence on data. The DGA strategy consists of three main parts. The first part involves an attention mechanism that assigns decoupled weights to the horizontal, vertical, and spectral dimensions. The second part decouples the coupling relationship between the three dimensions by adding the three values obtained by multiplying two of the three weights and improves the gaze effect by increasing the weight coefficients. The third part complements the weight coefficient results obtained in the second part by matching the weight features and assigning weights to the three types of weights. The utilization of multidimensional differences aims to differentiate the detailed differences between objects by leveraging the differences in each dimension. By assigning weights on different dimensions, useful information can be extracted from subtle feature differences. In summary, the DGA algorithm simulates the observation and gaze mechanism of the human eye, utilizes multidimensional feature information extraction, and reduces the dependence on data volume. This algorithm effectively extracts features from limited data and accurately distinguishes the detailed differences between objects.

2) REM: Deep learning methods often overlook the contextual relationships between pixels in HSIs, leading to noisy and confused categories in classification results. This phenomenon can be alleviated by reducing the redundancy and irrelevant information in HSIs. The key to reducing redundant information lies in extracting important feature information, such as texture, shape, and semantics, and finding the accurate contextual relationships between pixels while ignoring the irrelevant parts in feature maps. Therefore, it is crucial to remove unnecessary features while preserving the important ones to effectively enhance the model's anti-interference capability. The algorithm employs different components and activation functions, including Swish, tanh, and redundancy elimination operators. The Swish activation function with its smooth curve shape retains the activation values of useful features while suppressing the impact of redundant features. The nonlinear characteristics of the Swish function help the model better differentiate the differences between features and pay more attention to useful ones. Hence, the first branch of the algorithm utilizes the Swish activation function to minimize the influence of redundant features and enhance the model's feature selection capability.

3) PFM: HSIs contain a significant amount of detailed information, a large portion of which is redundant. Directly using images for deep network feature extraction is susceptible to interference from redundant information, resulting in incomplete network training and even training biases, ultimately leading to insufficient feature descriptive ability and poor image classification performance. PFM addresses this issue by separating parallel convolutions of multiple different scales during both the training and inference processes. These convolutions are designed to extract features related to texture

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and contour structures, enabling the discrimination of features in HSIs and improving the feature descriptive ability. This algorithm can extract more feature information from limited HSIs data and perform more accurate classification of features for different categories, thus reducing the dependence on data volume.

B. Low Data Dependence

Manual labeling of hyperspectral photographs at the pixel level is labor-intensive and time-consuming, especially when the dataset is huge and contains tens of millions of pixels. As a result, researching methods to lessen the reliance on deep learning models for HSI classification on data has significant practical implications for boosting the use of hyperspectral technologies. In general, neural networks require a huge amount of data for feature learning in order to modify parameters appropriately. The experimental results of this work show that the strategy that performs well with a given amount of training set partitioning does not perform well with a small number of training sample partitioning conditions. The experimental results on the Indian Pines, Salinas, DFC2018 Houston, and WHUHi-HongHu datasets reveal that they will cause a series of problems with class confusion and excessive noise, regardless of whether the scope of the dataset is small or large. This has a significant impact on the categorization effect of hyperspectral pictures. Although DGPF-RENet introduces slightly more noise than other approaches in some circumstances, it employs vision features and is more user-friendly than other methods, particularly for categories with small training sample sizes. In the case of partitioning low training data, the classification accuracy of categories with only one or several training samples is significantly improved, ensuring that the classification of the entire dataset is generally clear and closer to the ground truth and still performs well in the case of low training data with low data dependence.

C. Fast Convergence

The network's rapid convergence can help to assure training stability and reduce computer resource usage during the training process. To demonstrate the proposed approach's fast convergence capabilities, the Indian Pines 5% training process is used, and the verification set accuracy curves of the proposed technique and the eight neural network methods compared are generated, as shown in Fig. 13.

The figure shows that the DGPF-RENet model reaches convergence the quickest. In this study, we fixed the training epochs at 64 to allow for a fair comparison with other models and guarantee that other models also achieve a specific level of convergence. In fact, DGPF-RENet may produce comparable results even when epochs are not set to such high levels, demonstrating that this feature of its fast convergence is beneficial for enhancing model stability during training and minimizing the use of computer resources.

D. Possible Deficiencies

The main goal of this research is to construct a minimal data-dependent deep neural network that can adapt to applications requiring HSI categorization with limited data. Through

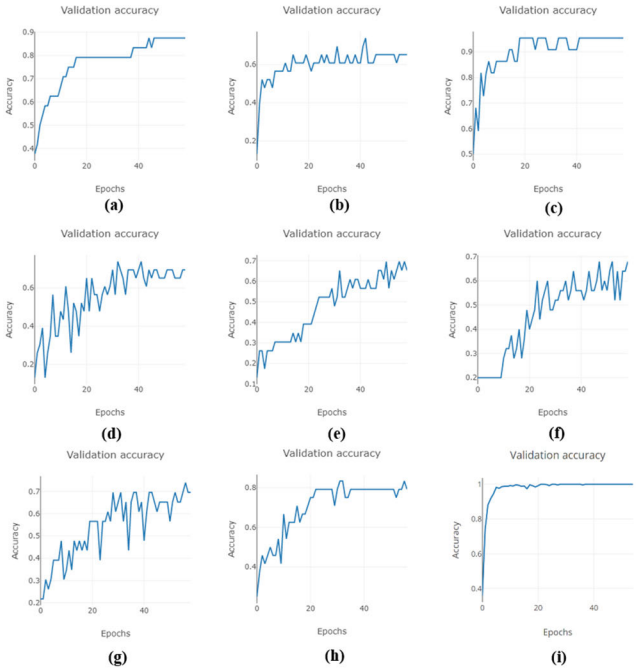


Fig. 13. Verification accuracy of the nine models reflects to a certain extent the speed of the model during the training process: (a) 2-D-CNN, (b) 3-D-CNN, (c) RCNN, (d) Yang, (e) ViT, (f) LeViT, (g) RvT, (h) HiT, and (i) DGPF-RENet.

the horizontal, vertical, and spectral dimension decoupled-gaze mechanism, DGA improves its learning capacity for less training samples, whereas REM eliminates any redundant information that might be present on dozens or hundreds of spectral dimensions. Comparing this to datasets with a lot of bands has a big advantage. However, some hyperspectral datasets have a limited number of bands, which somewhat impairs the actual performance of DGA and REM. DGPF-RENet, for instance, does not exhibit an absolute advantage in the DFC2018 Houston dataset with just 48 bands.

It is important to note that, for certain ground objects, even small amounts of training data may be difficult to obtain, which may limit practical feasibility. This leaves us to consider nonparametric or unsupervised methods. Recent studies have focused on metalearning to help the model understand the features of HSIs, so as to complete cross-domain recognition of data.

In addition, the classification process of DGPF-RENet proposed in this article can be regarded as a preprocessing step. The processing of HSIs in real application scenarios also includes processing steps, such as subpixel localization, spectral unmixing, and anomaly detection. However, this is beyond the research field of this article. In the whole process of hyperspectral data processing, the method in this article only is an important part.

VI. Conclusion

The application of deep learning methods to hyperspectral ground object classification has become increasingly widespread. The HSI ground object classification method DGPF-RENet, based on the DGA mechanism, the REM, and

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the PFM, has the advantage of its applicability to a small number of tagged samples and few training iterations. First, our proposed DGA decouples 3-D vectors into horizontal, vertical, and spectral dimension directions, then assigns coordination weights, and cleverly utilizes the gaze property of human vision to focus on the local differences of pixels. Next, REM is proposed to remove redundant information from the feature map and retain interpixel contextual information. Finally, PFM is proposed to enhance the learning ability of the model by boosting the convolutional scale. The experimental results show that there is low category confusion in the results of DGPF-RENet classification, and it has a strong ability to recognize objects in HSIs. Compared with nine other deep learning methods proposed in recent years for HSIs, the DGPF-RENet proposed in this article is more effective. Due to the complexity of HSIs capture, it might be challenging to collect even a tiny amount of data in certain scenarios. In the future, we plan to tackle this issue by introducing cross-domain paradigms, such as metalearning, to deal with the volatile and difficult-to-obtain data in natural scenarios.

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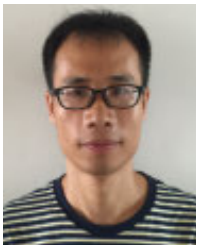
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